

Synthetic Text Generation for Training Large Language Models via Gradient Matching

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Abstract

Synthetic data has the potential to improve the performance, training efficiency, and privacy of real training examples. Nevertheless, existing approaches for synthetic text generation are mostly heuristics and cannot generate human-readable text without compromising the privacy of real data, or provide performance guarantees for training Large Language Models (LLMs). In this work, we propose the first theoretically rigorous approach for generating synthetic human-readable text that provides convergence, performance, and privacy guarantees for fine-tuning LLMs on a target task. To do so, we leverage Alternating Direction Method of Multipliers (ADMM) that iteratively optimizes the embeddings of synthetic examples to match the noisy gradient of the target training or validation data, and maps them to a sequence of text tokens with low perplexity. In doing so, the generated synthetic text guarantees convergence of the model to a close neighborhood of the solution obtained by fine-tuning on real data and preserves their privacy. Experiments on various classification tasks confirm the effectiveness of our proposed approach. Our code is available at <https://github.com/BigML-CS-UCLA/GRADMM>.

1. Introduction

High-quality data is crucial for training Large Language Models (LLMs) to superior performance (Yang et al., 2024; Li et al., 2023b). However, collecting and curating high-quality data is often very expensive and hard to obtain in many domains. In addition, as LLMs can memorize their

training data (Hartmann et al., 2023), ensuring the privacy of training examples hinders training the model directly on the training data. Thus, generating small subsets of synthetic data that can train an LLM to superior performance on the target task becomes handy. To do so, synthetic text should be generated in a way that ensures similar dynamics to that of training on the real data. However, text is discrete in nature and optimization in the discrete space is very challenging.

Existing approaches for synthetic text generation mostly rely on advanced LLMs such as GPT-4 to generate synthetic text for the target categories (Ye et al., 2022; Meng et al., 2022; Li et al., 2023b; Gupta et al., 2023; Tao et al., 2024; Wu et al., 2024; Dekoninck et al., 2024; Yu et al., 2024). LLM-generated text either suffers from lack of diversity and faithfulness to real data (Ye et al., 2022; Meng et al., 2022; Li et al., 2023b), or requires meticulous prompt engineering and highly complex pipelines, such as multi-agent frameworks, iterative sampling, and processing mechanisms (Gupta et al., 2023; Dekoninck et al., 2024; Wu et al., 2024). The complexity of the pipelines, efforts for manual prompt engineering, and the cost of querying advanced models limits the applicability of such approaches. A few recent studies explored the use of VAEs and diffusion for controllable text generation (Li et al., 2022; Gong et al., 2022; Zhou et al., 2024). But, training diffusion models is computationally heavy and difficult in practice. Importantly, none of the above approaches guarantees performance of LLMs trained on the synthetic text or preserve privacy of real data.

The above limitations raise a key question: *Can we generate a small subset of synthetic text that can train an LLM with similar dynamics to that of real data?* For vision models, Dataset Distillation (DD) addresses the above question by generating a small number of synthetic images that minimize the training loss (Wang et al., 2018; Loo et al., 2022; Nguyen et al., 2020), match the training gradient (Zhao et al., 2020; Zhao & Bilen, 2021) or model’s weight trajectory during training (Cazenavette et al., 2022; Wang et al., 2022). For images, gradient-based methods can easily operate in the pixel-wise continuous space. However, for LLMs, the discrete nature of text and the very large number of LLM’s parameters make DD much more challenging. The few ex-

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isting approaches generate synthetic embeddings that minimize the training loss (Sucholutsky & Schonlau, 2021; Li & Li, 2021; Sahni & Patel, 2023; Maekawa et al., 2023), or by training a generator model to match the gradient of an LLM trained on target data (Maekawa et al., 2024). However, the synthetic embeddings are not readable and cannot be transferred to train other LLMs, and synthetic data generated by matching dynamics of a LLM trained on target data may include real training examples and is not privacy preserving.

In this work, we propose the first theoretically-rigorous method to generate readable synthetic text that guarantees similar dynamics to that of fine-tuning on real data. First, we formulate a discrete optimization problem to find text embeddings that have a similar gradient to that of real data, under the constraint that the optimized embeddings should correspond to tokens in the vocabulary. Moreover, to ensure readability, we add another constraint that requires the sequence to have a low perplexity. Then, we solve this discrete optimization problem using Alternating Direction Method of Multipliers (ADMM) that iteratively optimizes the embeddings of synthetic data to match the average gradient of the real data, and maps them to a sequence of text tokens with low perplexity. To guarantee Differential Privacy (DP), we clip real data gradients and add controlled noise to their average before matching it. We prove that the synthetic text generated by our method guarantees convergence to a close neighborhood of the solution obtained by fine-tuning the model on real data.

We conduct extensive experiments to evaluate the effectiveness of our approach, namely GRADMM, for generating synthetic data using Phi model for multiple classification tasks. First, we consider the case where only a small number of validation examples are available and we apply GRADMM to generate a larger fine-tuning data. We show that with only 5 to 50 examples, GRADMM can successfully generate 100 synthetic data that outperform training on the real examples by up to 31.5%. Next, we apply GRADMM to generate a small synthetic data based on an existing larger fine-tuning data. We show that the synthetic data generated by GRADMM outperforms zero-shot and few-shot generation by LLMs as well as real examples selected by coreset selection methods by up to 13.1%, while ensuring the privacy of the training data. We also confirm the transferability of GRADMM’s generated text via Phi for fine-tuning other LLMs, including Llama-3.2-1B and OPT-1.3B.

2. Related Work

2.1. Dataset Distillation (DD)

DD aims to generate a small synthetic subset of examples that can achieve a similar generalization performance to that of training on the full real dataset.

DD for Images. DD is originally proposed for images. Wang et al. (2018) initially proposed a meta-learning approach which synthesizes data by iteratively training a model to convergence on the synthetic examples, and optimizing the synthetic data such that the trained model generalizes well on the real training data. Subsequent studies tried to make this process more efficient by using kernel methods to approximate training the model on synthetic data in a closed form (Loo et al., 2022; Nguyen et al., 2020). More recent works generate synthetic data by matching the gradient (Zhao et al., 2020; Zhao & Bilen, 2021; Kim et al., 2022) or wright trajectory (Cazenavette et al., 2022; Wang et al., 2022) of the model trained on real data, or by matching the data distribution (Zhao & Bilen, 2023).

DD for Text. There have been recent efforts in applying DD to text. For text datasets, existing methods (Sucholutsky & Schonlau, 2021; Li & Li, 2021; Sahni & Patel, 2023) apply the original meta-learning based method of (Wang et al., 2018), or minimize the KL-divergence between the self-attention probabilities of the model and the distilled attention labels across all layers and heads, for the first token (Maekawa et al. (2023)). As generating text in the discrete space is difficult, the synthetic data is generated as continuous input word embeddings instead of discrete text. Such embeddings cannot be used for training other models that have different word embedding weights, and are unreadable to humans, making them difficult to interpret and analyze. Sucholutsky & Schonlau (2021); Sahni & Patel (2023) transformed their distilled synthetic samples to text by finding words with the nearest neighbor embeddings. However, this results in unrelated words that are not meaningful.

To generate readable text, Maekawa et al. (2024) first trains a proxy language model from scratch to generate synthetic training data for different classes. Then, it fine-tunes a generator model to generate synthetic data by minimizing the gradient matching loss between generated and training data. Training the proxy model is a bottleneck in scaling the method. Besides, as the distilled synthetic data may include real samples from the original dataset, this method cannot ensure privacy.

Notably, none of the existing DD methods scale beyond BERT (Devlin, 2018) to LLMs with billions of parameters. In this work, we propose the first DD method that can generate privacy-preserving human-readable text, by matching gradients of LLMs with billions of parameters.

2.2. Synthetic Text Generation using Generative Models

LLMs. A large body of recent work used LLMs to generate synthetic text data in the zero-shot or few shot setting (Meng et al., 2022; Li et al., 2023b). In the zero-shot setting, the LLM is directly prompted to generate text for categories of interests. In the few-shot setting, a few

real-world data instances are provided as examples to guide the LLM in generating the synthetic data. In our work, we use the zero-shot and few-shot approaches as our baselines. LLM-generated text is often very repetitive and lacks diversity (Holtzman et al., 2019; Keskar et al., 2019). Besides, it does not capture the distribution of the target task and may contain incorrect or hallucinated examples (Ye et al., 2022; Meng et al., 2022; Gupta et al., 2023; Li et al., 2023b; Wu et al., 2023). To address these issues, recent methods rely on extensive prompt engineering to inject semantic diversity for each target category (Gupta et al., 2023) and design highly complex pipelines, such as model arithmetic which composes and biases multiple LLMs (Dekoninck et al., 2024), multi-step meticulous prompt engineering to inject domain knowledge, iterative sampling, and self-correction to rectify inaccurately labeled instances (Gupta et al., 2023), and retrieval-augmented generation techniques (Wu et al., 2023). Such pipelines require a large number of queries to advanced LLMs such as GPT-4 (OpenAI, 2023) and Claude3-Opus (Anthropic, 2023). This incurs a large financial cost and makes such approaches difficult to apply in practice. While synthetic data generated by LLMs are human-readable, LLMs may memorize and generate their training data (Hartmann et al., 2023). Hence, the synthetic data generated by LLMs do not preserve the privacy of their training data. Besides, it does not provide any theoretical guarantee for the performance of LLMs trained on them.

VAE and Diffusion. A few recent studies explored the use of VAEs and diffusion for controllable text generation (Li et al., 2022; Gong et al., 2022; Zhou et al., 2024). Such approaches train a diffusion model from scratch and parameterize structural and semantic controls by different classifiers and update the latent variables to satisfy the controls (Li & Li, 2021), or to add noise and denoise embeddings of real data (Zhao et al., 2020). This process is computationally very heavy and difficult in practice. Similar to LLMs, synthetic data generated by VAEs and diffusion models do not provide guarantee for the performance of the trained model.

3. Problem Formulation

Here, we formalize the problem of generating small human-readable synthetic text that can fine-tune an LLM with similar dynamics to that of real data. We also discuss two common use cases where such synthetic data is useful.

Setting. Consider a pretrained LLM with parameters θ and vocabulary $V = \{v_1, \dots, v_{|V|}\}$ containing all the words it has been trained to recognize and use. Consider a supervised fine-tuning dataset $\mathcal{D}_T = \{s^i\}$, where each example $s^i = (\mathbf{p}^i, \mathbf{r}^i)$ is a pair of prompt $\mathbf{p}^i$ and response $\mathbf{r}^i$ containing words in the vocabulary. The negative log likelihood loss is defined as $\ell(s^i, \theta) = -\log(\mathbf{r}^i | \mathbf{p}^i)$. The fine-tuning objective is thus to minimize the negative log likelihood loss

over the whole dataset $\mathcal{D}$ as $\ell(\mathcal{D}, \theta) = \frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \ell(s^i, \theta)$.

Problem formulation. Given a subsets of real examples from the fine-tuning data $\mathcal{D}_{\text{real}} \subset \mathcal{D}_T$, our goal is to generate synthetic data $\mathcal{D}_{\text{syn}} = \{q^i\}_{i=1}^k, q^i \notin \mathcal{D}_T \forall i$, containing r synthetic examples that do not belong to $\mathcal{D}_T$, such that fine-tuning the model on $\mathcal{D}_{\text{syn}}$ minimizes the loss on $\mathcal{D}_{\text{real}}$. Formally,

$$\arg \min_{\mathcal{D}_{\text{syn}}, |\mathcal{D}_{\text{syn}}| \leq r} \ell(\mathcal{D}_{\text{real}}, \theta^*), \quad \text{s.t.} \quad \theta^* \in \arg \min_{\theta} \ell(\mathcal{D}_{\text{syn}}, \theta). \quad (1)$$

Readability constraint. Importantly, we want the synthetic data to be human-readable. That is we want every synthetic example to be a sequence of words in the vocabulary. Besides, to ensure that the sequence is meaningful, we require that the synthetic data has low perplexity. Thus, we wish to solve the following constrained optimization problem:

$$\arg \min_{\substack{\mathcal{D}_{\text{syn}}, |\mathcal{D}_{\text{syn}}| \leq k, \\ s \in \Gamma, \text{ppl}(s) \leq \epsilon \\ \forall s \in \mathcal{D}_{\text{syn}}}} \ell(\mathcal{D}_{\text{real}}, \theta^*), \quad \text{s.t.} \quad \theta^* \in \arg \min_{\theta} \ell(\mathcal{D}_{\text{syn}}, \theta), \quad (2)$$

where $\Gamma = \{s = (\mathbf{p}, \mathbf{r}) | p_j, r_j \in V\}$ is the set of all prompts and responses that consist of words in vocabulary V .

Use cases for synthetic data generations. The above formulation is applicable to two settings: (1) Data is scarce for the target task, and we want to generate a larger synthetic fine-tuning data based on a small number of examples from the target task. (2) A relatively large supervised fine-tuning data is available, and we wish to generate a smaller synthetic data to replace the real data to preserve the privacy of training examples or to improve the training efficiency.

4. Method

Next, we discuss our proposed method for generating readable synthetic text for fine-tuning LLMs on a target task.

4.1. Text Generation via Gradient Matching

An effective way to solve Eq 2 is via gradient matching. Specifically, we generate a synthetic data $\mathcal{D}_{\text{syn}}$ that has a similar gradient to that of the real dataset:

$$\arg \min_{\substack{\mathcal{D}_{\text{syn}}, |\mathcal{D}_{\text{syn}}| \leq r, \\ s \in \Gamma, \text{ppl}(s) \leq \epsilon \\ \forall s \in \mathcal{D}_{\text{syn}}}} D(\nabla_{\theta} \ell(\mathcal{D}_{\text{syn}}, \theta), \nabla_{\theta} \ell(\mathcal{D}_{\text{real}}, \theta)). \quad (3)$$

where $D(\cdot, \cdot)$ is a distance between two gradients. Following (Deng et al., 2021; Geiping et al., 2020), we use $1 - \cos(\cdot, \cdot)$ as our distance metric, where $\cos$ is the cosine similarity. If such a synthetic data can be generated, training on it with gradient methods directly minimizes the loss on real data.